

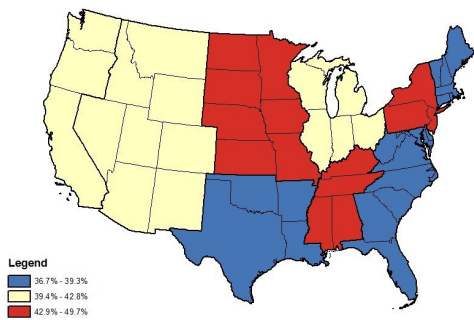


Figure 1: Regional Pancreatotomy rates for early-stage pancreas cancer; tertiles

Mo1350

PARENCHYMAL-PRESERVING RESECTION FOR PANCREATIC NEUROENDOCRINE TUMORS: ARE THE OUTCOMES JUSTIFIED?

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OBJECTIVE: To evaluate the outcomes of segmental pancreatectomy (SP) and enucleation (EN) in the management of low- and intermediate-grade pancreatic neuroendocrine tumors (PNETs). **BACKGROUND:** The extent of resection needed for PNETs is debated. The parenchymal-preserving options of SP and EN may be an alternative to more extensive resections. **DESIGN:** Academic Center. Retrospective review of single-surgeon experience. **RESULTS:** Forty-two patients underwent either SP (n=17) or EN (n=25) for PNETs from 2002 to 2016. Mean tumor size was 1.9 ± 1.22 cm (0.7-5.3 cm) and 15 were ≥ 2 cm. Thirty-three tumors were nonfunctional and 32 were low-grade. EN was usually performed for head lesions, and was completed laparoscopically/robotically in 17 patients (68%). Twenty-one patients underwent lymphadenectomy, with a mean lymph node count of 3.8 (1-16). Three patients (7%) had lymph node metastasis. There was no mortality. Significant post-operative morbidity occurred in 24%, and mean and median length of stay was 7.2 and 6.5 days. Grade B/C pancreatic fistulae occurred in 6 patients (14%). After a mean and median follow-up of 42 and 30 months, no patient has had tumor recurrence. There were 3 cases of de novo diabetes (7%) and 2 of exocrine insufficiency (5%). New or worsened pancreatic atrophy and/or duct dilation was demonstrated in 21% of patients on imaging follow-up. Development of pancreatic insufficiency was significantly associated with post-operative pancreatic atrophy (p=0.0001). **CONCLUSION:** Parenchymal-preserving resections offer a safe and effective management option for PNETs, with low recurrence rates and excellent intermediate outcomes. Use of minimally invasive techniques and application to tumors ≥ 2 cm represent important advancements. Longer-term follow-up is needed, particularly for patients showing pancreatic atrophy.

Mo1351

AMPULLARY STENOSIS AFTER ROUX-EN-Y GASTRIC BYPASS SURGERY: A NEW PHENOMENON IN THE POST-BARIATRIC SURGERY POPULATION

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Introduction: Roux-en-Y gastric bypass (RYGB) surgery is routinely performed for weight loss in morbidly obese patients. At our institution, we identified a small number of patients who developed benign ampullary stenosis after previously undergoing RYGB. **Cases:** Nine patients (4 male, 5 female) with mean age 56.3±7.6 years old at presentation, had previously undergone RYGB for morbid obesity a mean 8.9±4.8 years prior to diagnosis of ampullary stenosis (Table 1). All patients had obstructive liver function test patterns and either MRCP, ERCP, or cholangiogram confirming ampullary stenosis (Figure 1). The presenting complaints/diagnoses included: abdominal pain (4 patients), back pain (1 patient), incidental finding of elevated liver enzymes (1 patient), primary sclerosing cholangitis (1 patient), and ascending cholangitis (4 patients). 5 patients had cholecystectomy prior to presentation. 2 patients (1 male, 1 female) had comorbidities of systemic lupus erythematosus (SLE), and 3 patients had hypothyroidism. 3 patients had positive anti-nuclear antibody titers, which included both patients with SLE. **Management and Outcomes:** A variety of interventions were performed to treat the ampullary stenosis. 2 patients received transduodenal ampullectomies. 2 patients received choledochoduodenostomies (one with common duct stone removal). 1 patient underwent laparoscopic assisted transgastric ERCP with sphincterotomy and common duct stone extraction. 1 patient was managed only with percutaneous transhepatic biliary drainage. 1 patient, in whom pre-operative ampullary malignancy was suspected, underwent a pylorus sparing Whipple procedure. 1 patient underwent laparoscopic cholecystectomy as an initial diagnostic maneuver for abdominal pain. 1 patient, thought to have a diagnosis of primary sclerosing cholangitis, underwent liver transplant and died from postoperative complications. Together, these interventions yielded 3 ampulla specimens which were all negative for malignancy after pathologic analysis. All other hepatobiliary and cytopathology specimens were negative for any evidence of malignancy. **Discussion:** Nine cases of benign ampullary stenosis affecting patients who have undergone RYGB are presented. We postulate that metabolic or hormonal derangements after RYGB alter ampullary physiology causing dysfunction or benign stricture formation. Another possibility is an autoimmune mechanism since several patients carried diagnoses of SLE, primary sclerosing cholangitis, and hypothyroidism. We present this case series to raise attention to the entity of benign ampullary stenosis associated with RYGB. Additional mechanistic studies are necessary to further understand the pathophysiology of this peculiar clinical phenomenon.

Patient Characteristics

Patient	Age at Diagnosis	Sex	Year of RYGB	Presenting Symptoms/Diagnosis	Intervention and Year Performed	Relevant Pathology	Past Medical History	Notable Labs
1	54	Female	2012	Epigastric pain, nausea	Transduodenal ampullectomy with sphincteroplasty, 2014	NA	Hypothyroidism, nephrolithiasis, atrial fibrillation, hypertension, anxiety, depression	ANA positive
2	50	Male	2006	Primary sclerosing cholangitis	Liver transplant, 2015 (Deceased postoperatively)	Liver biopsy: biliary obstruction, mild steatosis	GERD, hypertension, anemia	ANA positive
3	58	Female	2010	Right upper quadrant pain, cholangitis	Percutaneous transhepatic biliary drain, 2014	Benign biliary brushings	Heavy alcohol use	NA
4	73	Female	2009	Weight loss, back pain, slightly elevated CA 19-9	Pylorus sparing Whipple procedure, 2015	Pre-op biliary brushings: malignant cytology. Ampulla (operative specimen): periampullary ulceration and fibrosis with sub-mucosal abscess. No tumor. Focal increase of plasma cells in periductal infiltrate, rare IgG4 cells.	Systemic lupus erythematosus, aortic stenosis	Pre-op CA 19-9: 37 (reference <36)
5	51	Male	2005	Incidentally elevated liver function enzymes	Transduodenal ampullectomy with choledochoduodenostomy, 2016	Ampulla: No neoplasm identified	Systemic lupus erythematosus, Raynaud's syndrome, nephrolithiasis, hepatitis C, fatty liver	ANA positive
6	53	Male	2005	Epigastric pain, cholangitis	Open cholecystectomy and choledochoduodenostomy, 2016	Common bile duct margin: no high grade dysplasia or carcinoma	Dilated viral cardiomyopathy, heavy alcohol use	NA
7	63	Male	2007	Cholangitis	Laparoscopic-assisted transgastric ERCP with sphincterotomy and common duct stone extraction, 2016	Stomach (gastrostomy site): gastric tissue with serosal fibrous adhesions	Hepatitis C, hypertension, COPD, chronic pain	NA
8	49	Female	1997	Cholangitis	Common bile duct exploration, common duct stone extraction, choledochoduodenostomy, 2016	NA	Hypothyroidism, fibromyalgia, migraines, endometriosis, obstructive sleep apnea, GERD, PTSD	NA
9	56	Female	2006	Abdominal pain, nausea, emesis	Laparoscopic cholecystectomy, 2016	Gallbladder: mild chronic cholecystitis. No calculi present.	Hypothyroidism, asthma, GERD, anxiety, depression	NA



Figure 1: Cholangiograms (upper row and bottom left) and trans-gastric ERCP (bottom right) demonstrating ampullary stenosis.

Mo1352

CHARACTERISTICS OF ABDOMINAL SYMPTOMS IN POST LAPAROSCOPIC CHOLECYSTECTOMY PATIENTS: A MULTI-CENTERED CROSS-SECTIONAL STUDY IN SHANGHAI

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Background and Objective Patients experienced laparoscopic cholecystectomy (LC) should have symptoms relief. However, some of the patients underwent LC suffered from distressing pain or other abdominal symptoms, which is defined as post-cholecystectomy syndrome. The prevalence and characteristics remain unclear. The aim of this study was to investigate the clinical characteristics and the occurrence rate of abdominal symptoms after LC in Chinese patients. **Methods** This multi-centered, cross-sectional study was conducted in five tertiary teaching hospitals in Shanghai from August to December in 2014. Outpatients (n=793) from General or Biliary Surgery Department between 18 to 80 years old who underwent LC at least 1 month before enrollment completed the Abdominal Symptom Questionnaire.